

Cluster 1

Cluster 1: HALLMARK EPITHELIAL MESENCHYMAL TRANSITION
 Cluster 1: HALLMARK TGF BETA SIGNALING
 Cluster 1: HALLMARK MITOTIC SPINDLE
 Cluster 1: HALLMARK APICAL JUNCTION
 Cluster 1: HALLMARK UV RESPONSE DN

Cluster 2

Cluster 2: HALLMARK HEME METABOLISM
 Cluster 2: HALLMARK MYOGENESIS
 Cluster 2: HALLMARK TNFA SIGNALING VIA NFKB
 Cluster 2: HALLMARK KRAS SIGNALING DN
 Cluster 2: HALLMARK INFLAMMATORY RESPONSE

Cluster 3

Cluster 3: HALLMARK CHOLESTEROL HOMEOSTASIS
 Cluster 3: HALLMARK ESTROGEN RESPONSE LATE
 Cluster 3: HALLMARK FATTY ACID METABOLISM
 Cluster 3: HALLMARK ESTROGEN RESPONSE EARLY
 Cluster 3: HALLMARK BILE ACID METABOLISM

Cluster 4

Cluster 4: HALLMARK MYC TARGETS V1
 Cluster 4: HALLMARK E2F TARGETS
 Cluster 4: HALLMARK G2M CHECKPOINT
 Cluster 4: HALLMARK MYC TARGETS V2
 Cluster 4: HALLMARK MTORC1 SIGNALING

Cluster 5

Cluster 5: HALLMARK ESTROGEN RESPONSE LATE
 Cluster 5: HALLMARK KRAS SIGNALING DN
 Cluster 5: HALLMARK ESTROGEN RESPONSE EARLY

0 10 20 30 40 50
 $-\log(p\text{-value})$

Significance



adjusted $p < 0.05$



nominal $p < 0.05$